Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B. Tech CE 7th semester
Name of the Paper: **Transportation Engineering-II (Railway and Tunnel)**
Paper Code: **TCE 702**
Time:1.5hour

Maximum Marks: 50

Note:
(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1. (10 Marks) CO1

a. Discuss the various modes of transportation and their importance in global and regional contexts. How does each mode contribute to the overall efficiency and sustainability of transportation systems?

OR

b. Analyze the role of Indian Railways in the universal transportation scenario. How has the development of railway tracks in India evolved over time, and what are the key factors driving modernization in this sector?

Q2. (10 Marks) CO1 & 2

a. Describe the component parts of a railway track, focusing on gauges, wheel, and axle arrangements.

OR

b. Explain the concept of resistance to traction in railway systems. What are the various types of resistances that affect train movement?

Q3. (10 Marks) CO1 &2

a. Define hauling capacity and tractive effort in railway engineering. Discuss the stresses experienced by key components like rails, sleepers, ballast, and track formations, and how these stresses are managed to ensure safe and efficient train operations.

OR

b. Explain the concept of coning of wheels in railway systems. How does it impact the stability and smooth movement of trains.

Q4. (10 Marks) CO2

a. What is the theory of long welded rails, and how does it improve railway track performance? Explain the benefits and challenges associated with the use of long welded rails in modern railway networks.

OR

b. Describe the different types of sleepers used in railways, along with their specific requirements.

Mid Term (Odd) Semester Examination October 2024

Q5. (10 Marks) CO2

a. Analyze the requirements and design specifications for ballast in railway tracks. How are rail-to-rail and rail-to-sleeper fittings implemented.

OR

b. Discuss the causes and effects of creep, wear, and joints in railway tracks.